



7292 - SuperDisks: A high-fidelity public database of protoplanetary and debris disk MIRI-MRS spectra

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Our understanding of the physics and chemistry of planet-forming regions in the inner parts of protoplanetary disks (< 1 au) is currently undergoing a revolution due to the incredible spectroscopic data returned from JWST-MIRI and NIRSpec. We propose SuperDisks, a legacy archival program to uniformly process every public MIRI-MRS observation of protoplanetary disks observed during JWST Cycles 1-3, and conduct the first complete demographic analysis of planet-forming disks across the stellar mass and age range, from 1 Myr to the debris disk stage at 100 Myr. Using the proven reduction process from the JDISCS collaboration and their dedicated asteroid calibrators, SuperDisks will yield ~ 240 high-quality, homogeneously reduced spectra from Cycles 1-3 of molecular and atomic emission lines, solid-state features, and spatially extended H₂ emission maps from disks. The data products will have significantly improved spectral fidelity of 0.3% (1 σ), spectro-photometric precision to 1%, and wavelength calibration better than 5 km/s across the MIRI-MRS range. We will further decompose components in multiple systems, where present, and recover observations that were mispointed or incomplete. We will use the database to search for global relations between molecular emission

luminosities and stellar and system parameters. All data will be delivered to MAST as high-level science products (HLSPs), as well as to the SpExoDisks database to enable the community to fully leverage MIRI spectroscopy of planet-forming disks.

OBSERVING DESCRIPTION